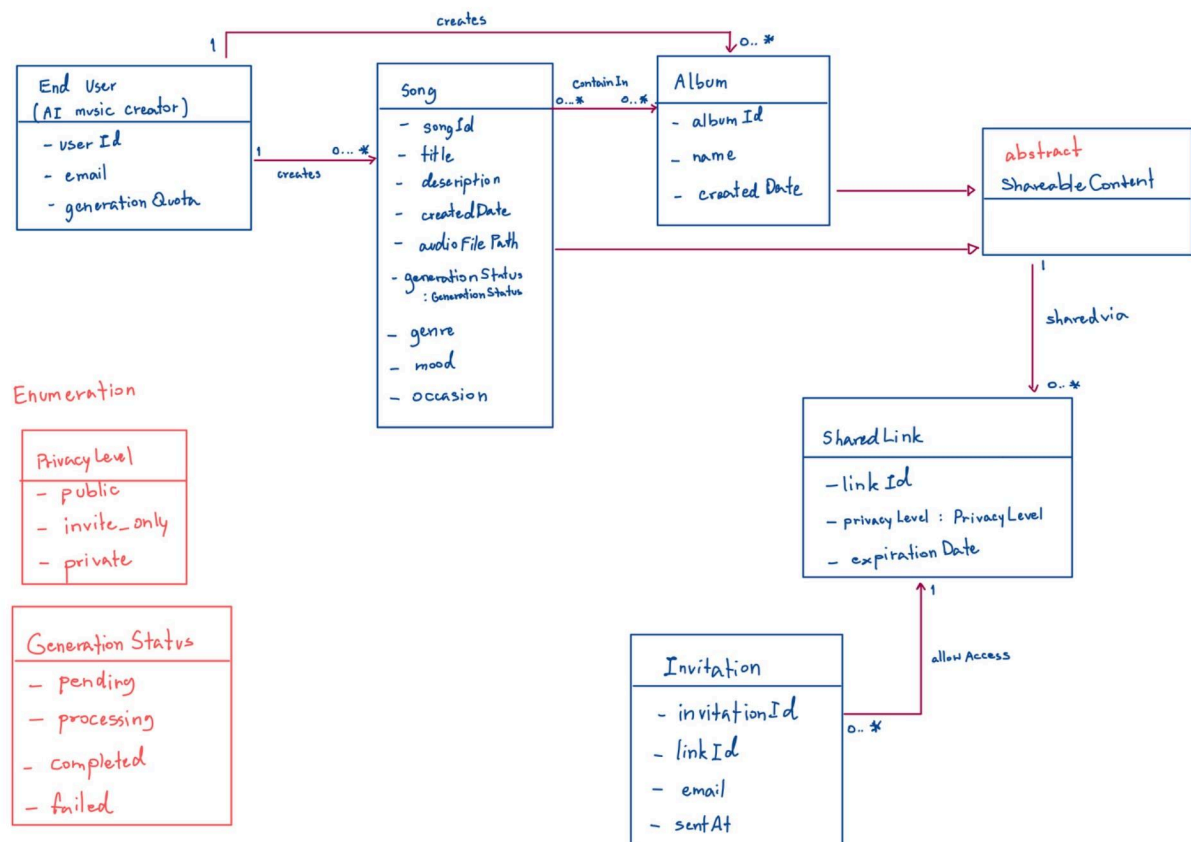


Domain Model Diagram



Domain Constraints

- Each Song must belong to exactly one User.
- A User may create zero or more Songs.
- A User may own zero or more Albums.
- A Song may exist in zero or more Albums.
- A Song may only be included in an Album owned by the same User.
- Access to shared content must follow the selected PrivacyLevel.
- Invite-only SharedLink must specify one or more authorized email addresses.
- A User may not generate Songs beyond the assigned generationQuota.
- Each Invitation must be associated with exactly one SharedLink.
- Each SharedLink must reference exactly one ShareableContent.

Assumptions

1. Each Song is owned by exactly one User.

Reasoning:

BR-02 states that generated songs are associated with the user who created them, but does not specify whether ownership can be shared. Therefore, single ownership is assumed to simplify access control and modification rights.

2. ShareableContent represents any resource that can be shared externally via a SharedLink.

Reasoning:

UC-04 indicates that both Songs and Albums can be shared using a generated link. Therefore, a common abstraction is assumed to generalize shareable resources in the domain model.

3. A ShareableContent may have multiple SharedLinks over time.

Reasoning:

The SRS does not restrict how many shareable links may be generated for a resource. Therefore, it is assumed that users may generate new links (e.g., after expiration), resulting in multiple SharedLinks referencing the same Song or Album.

4. A SharedLink may optionally have an expiration date.

Reasoning:

The SRS does not specify whether shareable links must expire or how expiration is handled. Therefore, expiration is assumed to be optional.

5. Songs may only be included in Albums owned by the same User.

Reasoning:

The SRS does not mention support for collaborative album management. Therefore, cross-user organization of Songs into Albums is assumed to be unsupported.

Traceability Mapping

Domain Class	Derived from SRS	Description
User	US-01, FR-01	Represents the AI Music Creator who generates, manages, and shares Songs within the system.
Song	UC-02, FR-10	Represents the generated music artifact that is stored, played, organized, and shared by Users.
Album	US-06, FR-12	Represents a collection of Songs created by a User for organizational purposes.
ShareableContent	UC-04, FR-20	Abstracts common sharing behavior between Song and Album, both of which can be shared via generated links.
SharedLink	UC-04, FR-20	Represents an access mechanism that allows Songs or Albums to be shared externally with controlled visibility.
Invitation	FR-21, NFR-13	Represents authorized email access for invite-only SharedLinks.